METHOD 17 / A COMPLETE BEGINNER GUIDE

Transfer entropy

Question this method answers: Does the past of series X add information about the next value of Y beyond what Y's own past already explains?

Transfer entropy $TE_{X \to Y}$ is measured in nats (natural log). It is directional: $TE_{X \to Y}$ need not equal $TE_{Y \to X}$.

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No reported TE scores from major-pair experiments — only symbolic teaching rows.

Prepared for: 17 - Transfer Entropy. README status: not started.

02 / THE BASIC IDEA

What is TE_{X→Y}?

Standard correlation and MI at the same time index are symmetric summaries. Transfer entropy looks one step ahead: it rewards cases where X's history helps forecast Y's next move after accounting for Y's history.

Not structural causation

TE > 0	X past adds predictive information about Y future in this model.
TE = 0	No added information in the discrete population case assumed.
Hidden confounders	Common drivers can inflate TE without a direct mechanism.

Schreiber (2000) framed TE as a nonparametric information flow measure — useful for hypothesis generation, not proof of economic causality.

FX percent math (majors)

```
Simple return = P_t / P_{t-1} - 1
Log return = ln(P_t / P_{t-1})
```

Example: 1.1000 → 1.1110 is +1% simple move. TE is built on an ordered return series: define Y_{t+1}, Y past lags, and X past lags on the same clock.

Reverse quote direction flips log-return signs — keep one convention per leg. Gaps shorten usable lag vectors.

04 / DESIGN CHOICES

Lags and embedding

Choose lag length k for Y past and m for X past before analysis. TE depends on those orders — document k, m, and any binning of returns.

Section 17 in forex-formulas-explained.pdf ties TE to the repo's correlation method list.

Align sessions so "next day" means the same thing for both series.

05 / SCHREIBER (2000)

Research notes

Measuring information transfer. Physical Review Letters, 85(2), 461–464. DOI: 10.1103/PhysRevLett.85.461.

Introduced transfer entropy to quantify directed information flow using conditional probabilities, linking to Granger-style prediction ideas in an information-theoretic form.

Later papers address bias correction, continuous estimators, and multivariate extensions — pick one pipeline and stay consistent across pairs.

Symbol dictionary

Symbol	Meaning
Y_{t+1}	Next value of target series Y.
Y_past	Vector of past Y values (lags).
X_past	Vector of past X values (lags).
TE_{X→Y}	I(Y_{t+1}; X_past, Y_past) - I(Y_{t+1}; Y_past) in nats.
I(·;·)	Mutual information with natural log.
I(·;·|·)	Conditional mutual information.

07 / DEFINITION USED IN PAPERS

Conditional MI form

```
TE_{X→Y} = I(Y_{t+1}; X_past | Y_past)
= I(Y_{t+1}; X_past, Y_past) - I(Y_{t+1}; Y_past)
```

The second line is the computation recipe: how much information X past adds when Y past is already known about the next Y.

Swap roles to get TE_{Y→X} — compare directions, do not average blindly.

08 / PREDICTIVE GAIN

Intuition with words

If Y is a random walk, its own past may already fix next-step uncertainty; X past adds nothing → TE_{X→Y} = 0 in the idealized model.

If X leads Y by one day in a discrete teaching chain, TE_{X→Y} can rise while same-time MI also rises — they answer different timing questions.

09 / ESTIMATION PATH

Discrete bins on returns

Bin each return into a small alphabet (e.g., down / flat / up). Build empirical counts for tuples (Y_{t+1}, Y_past bins, X_past bins). Plug into the information formulas with care for sparse cells.

Continuous TE estimators (k-NN, kernel) avoid bins but need more data and tuning — plan those in experiments, not in this beginner PDF.

Teaching tuple sketch

Each row is one time step with binned symbols (0/1) for illustration — not EURUSD prices.

Row	Y_{t+1}	Y_past	X_past
1	1	0	0
2	0	1	0
3	1	0	1
4	0	1	1

```
TE_X_to_Y (discrete teaching estimator) ~ 0.000000 nats
```

Positive value means X_past changed conditional uncertainty about Y_{t+1} in this toy table. Recompute only after changing bins or lag definition.

11 / REPO USAGE

Daily majors workflow

1. Fix lag orders and bin edges

Write k, m, and return type in NOTES before scanning pairs.

2. Report both directions

TE_{X→Y} and TE_{Y→X} can differ — especially under asymmetric information flow.

3. Surrogate tests

Shuffle or block-bootstrap X past to build null distributions when n allows — raw TE alone is hard to rank.

Experiments under experiments/v001/ should store TE settings beside static correlation columns.

Estimation notes

```
# te_from_bins in build_guide.py for the 4-row teaching table
# Production: dedicated TE libraries with bias correction
```

Check whether your package uses natural log nats or bits. Match units across methods 15 and 17 in the same report.

Sparse bins with small n inflate TE — merge bins or widen the return window before trusting three decimal places.

Save random seeds and surrogate counts when experiments add significance bands; this guide does not prescribe a single null recipe.

Limits

TE is not VAR Granger causality with linear Gaussian assumptions — related idea, different object. It is not proof that trading X predicts Y profitably after costs.

Microstructure noise, asynchronous quotes, and holiday calendars distort lag alignment. TE can reflect preprocessing as much as economics.

Practice

Q: If TE_{X→Y} > 0, did X cause Y? **A:** Not necessarily — confounding and reverse mechanisms remain possible.

Q: Is TE symmetric? **A:** No — compare both directions explicitly.

Q: Same-time MI vs TE? **A:** MI ignores ordered prediction; TE uses lags and a future step.

Q: Can you TE price levels? **A:** Usually prefer returns; levels trend and inflate apparent flow.

Sources

[1] Schreiber, T. (2000). Measuring information transfer. Physical Review Letters, 85(2), 461–464.
Original transfer entropy paper.
Open source

[2] forex-formulas-explained.pdf — section 17 (this repository).
Project formula sheet.
Open source

Rebuild: python build_guide.py. End of transfer entropy beginner guide.